

Identifying Influential papers in DBLP Dataset

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Introduction

This report presents our results of a comprehensive analysis of the DBLP dataset, which contains over three million papers published up to October 2017. Our goal for this data analysis is to determine what papers were the most influential in shaping research trends, and how can exploratory data analysis (EDA), clustering, and temporal classification be used to identify the determining factors that make a paper influential?

To address this goal, we applied four complementary analyses. First, we conducted exploratory data analysis to uncover patterns in areas such as publication volume and citation distributions. Second, we applied K-Means clustering to TF-IDF text features to group papers into research areas and identify which communities have driven citation volume. Third, we created a citation network to compute PageRank scores to measure the structural influence beyond raw citation counts. Finally, we trained temporal classification models on pre-2010 papers and evaluated their ability to predict paper influence on post-2010 papers, identifying the most predictive textual and structural features.

Pre-Processing

Before our data analysis could be conducted, we uploaded the database into Amazon's AWS S3 service and processed it through Github via a streaming pipeline designed to handle the dataset without loading all records into memory simultaneously. Each file is read line by line, with individual JSON objects accumulated into chunks of 50,000 before being cleaned and appended to the output. This keeps our memory usage in check regardless of the file size, but may affect our results.

We filter out each chunk using a series of rules. Firstly, any record with a missing ID, Title, or Year was dropped entirely, as we deemed these were the minimum required fields for any analysis. Records with a blank or whitespace-only title were also removed. Duplicate paper IDs were filtered out within each chunk as well.

Secondly, unlike our first rule, other fields such as abstract and venue were not part of our hard filter criteria. We replaced abstract and venue fields with empty strings so these would remain within the dataset. This would allow papers without abstracts to contribute their title to TF-IDF features. The venue field is retained for similar reasons for EDA visualizations, but is not used as a feature or label in other methods.

Other fields were also created that were not originally part of the dataset. "author_count" is composed as a length of authors listed for each record, and is defaulted to zero if the field is either absent or not a list. "reference_count" are computed directly from the references list. "text_combined" is produced by concatenating the title with the abstract using a single space as a separator, which creates the input string to be used in TF-IDF for both clustering and temporal classification pipelines. Finally, we casted the year field into a numeric type and dropped records where the conversion failed to ensure the year field was always an integer.

After this preprocessing is completed, we exported the cleaned dataset into a Parquet file. We configured our programs to download this cleaned Parquet file prior to executing to ensure preprocessing runs once and each program uses the same, consistent dataset across all analysis processes. We also generated a metadata summary to record the cleaned dataset as well, with columns such as total paper count, year range, unique venue count, and missing field counts.

Exploratory Data Analysis (EDA)

Data Completeness

Before we began, we analyzed the completeness of the dataset across its key fields: references, authors, venue, and abstract. The figure below displays the data audit.

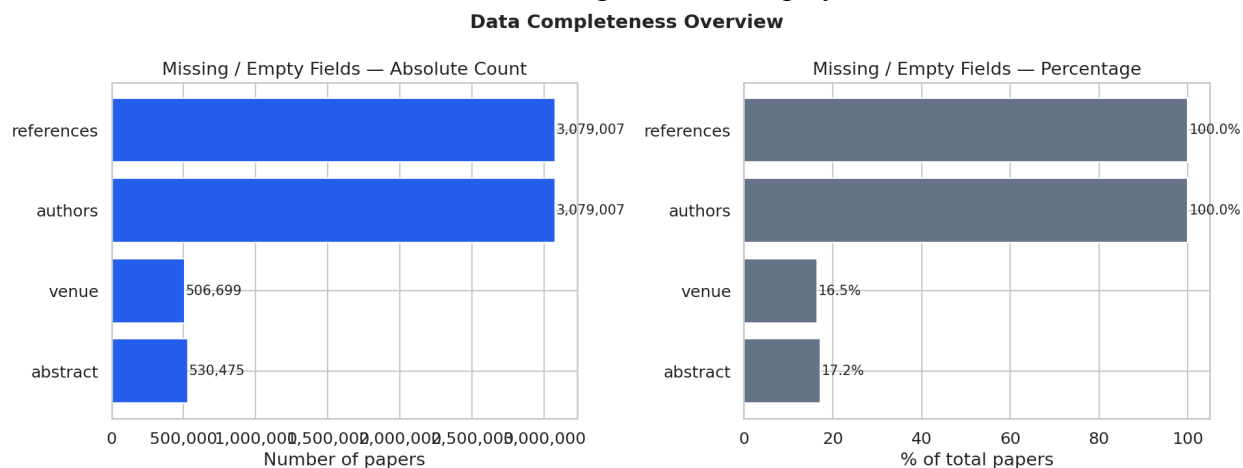


Figure 1: Data Completeness Overview - Missing/Empty fields by Count and Percentage

The most significant finding is that while the references and authors fields were present for all three million records, they were stored as either empty lists or had no data, which our completeness check section of our program flags as missing. Additionally, the venue field is missing for 506,699 (16.5%) of the papers, and the abstract field is missing for 530,475 (17.2%) of the papers as well. Instead of dropping these empty fields, we replace them with empty strings during preprocessing, such as the abstract field being utilized in TF-IDF in clustering and classification. As discussed in preprocessing, these empty abstract fields only contribute their title to the text_combined field. Similarly, the venue field is also only used in EDA to produce venue-level charts, and is not used in any class label or feature in any models.

Publication Volume

Figures 2 and 3 below examine how the volume of computer science research has grown over the decades covered by the dataset:

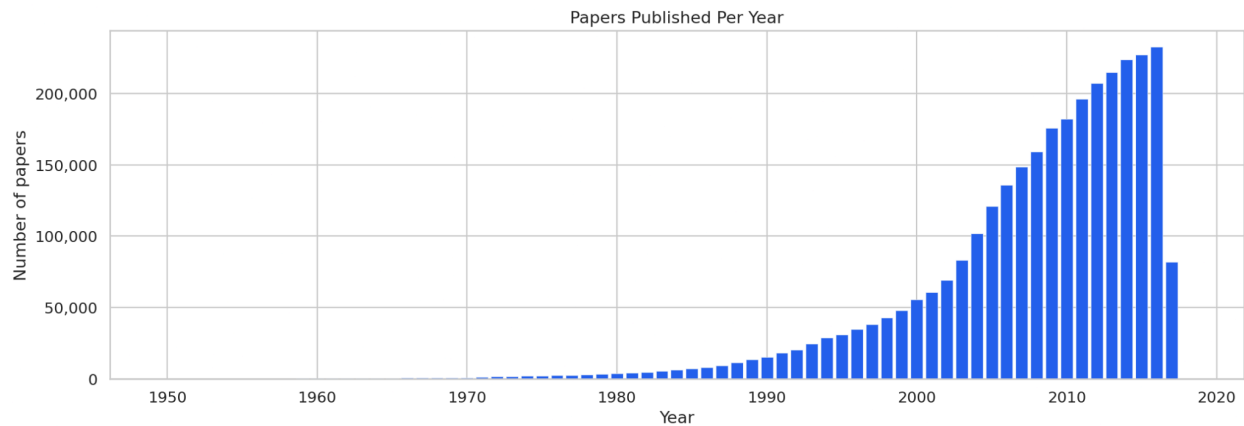


Figure 2: Papers Published per Year (1950-2020)

In figure 2, we can see there is almost negligible progress from 1960 to 1970, with steady growth from 1980 onward. One can also see that there is a significant increase after 2000, with annual paper counts reaching over 200,000 around 2014-2016 before sharply dropping around 2017. This sharp drop is most likely due to the dataset being cut off in October of 2017 rather than a decline in output. This consistent, upward trend suggests that computer science is an expanding discipline.

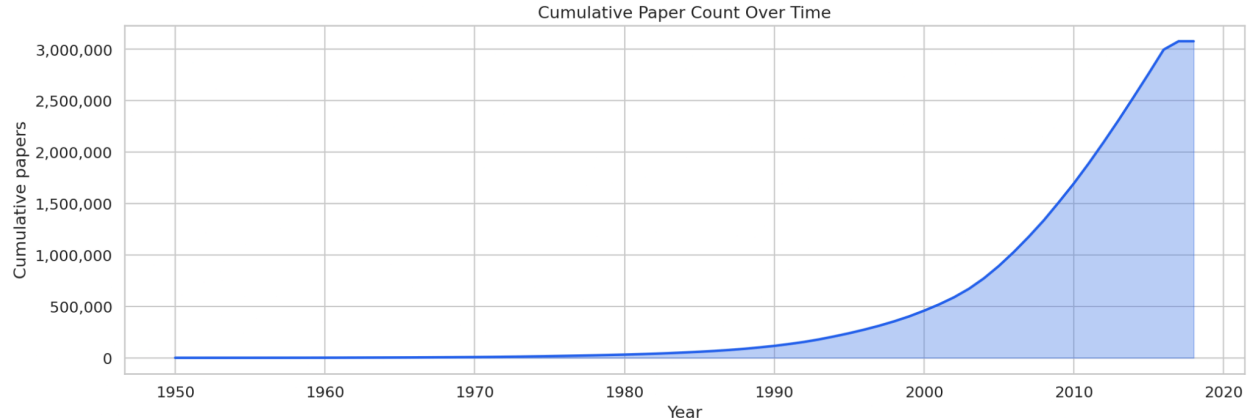


Figure 3: Cumulative Paper Count Over Time (1950-2020)

In figure 3, we can see the cumulative paper count curve that better reflects the bulk of the dataset in the recent years. Roughly half of the three million papers in the dataset were published after 2010, and over 80% were published after 2000. This has significant implications for temporal classification, as any model on pre-2010 data may not reflect data of post-2010 research.

Citation Distributions

We also used citations to determine which paper is considered “influential” in the dataset. Figures, 4, 5, and 6 all examine citation distributions from different angles and methods:

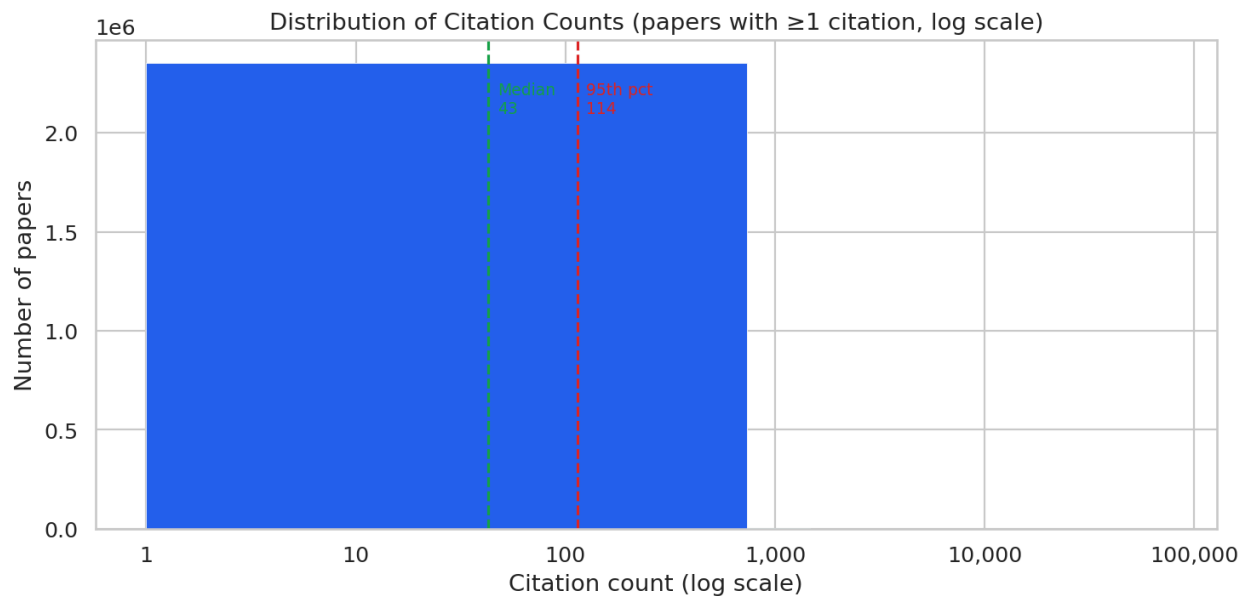


Figure 4: Distribution of Citation Counts (Papers with ≥ 1 citation, log scale) – Median 43, 95th Percentile 114

Among papers with at least one citation, the distribution is right skewed even on a log scale. From the graph, we can see the median is at 43, which means half of the cited papers have less than 43 citations, and the 95th percentile is at 114, meaning most of the citations are concentrated on a small number of papers.

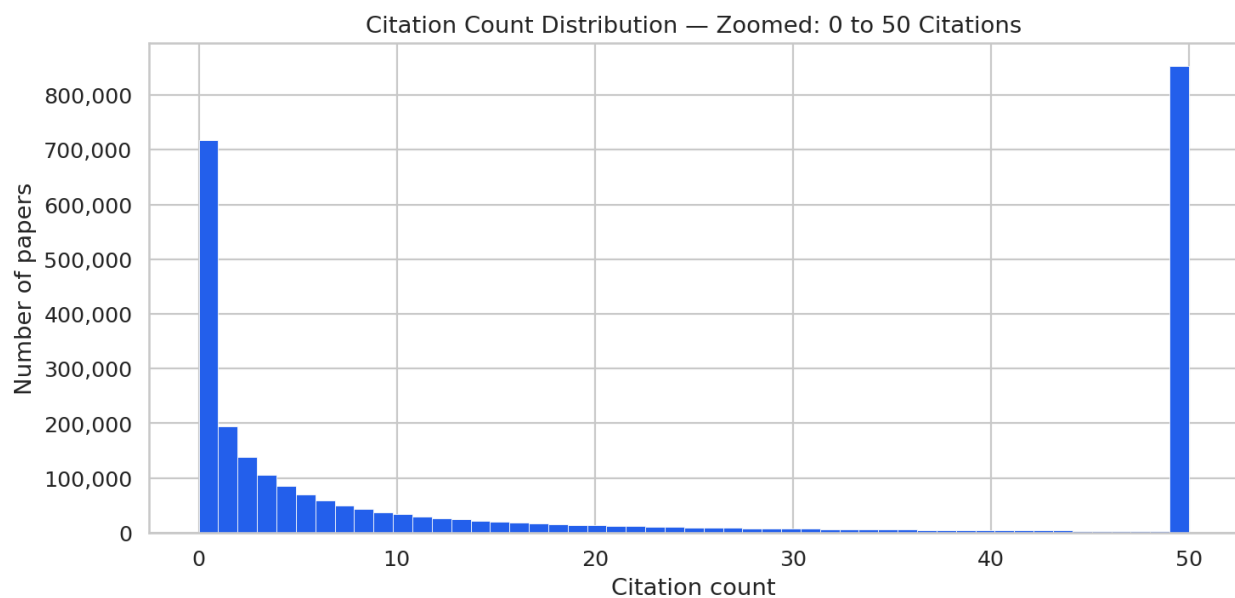


Figure 5: Citation Count Distribution (Zoomed: 0 - 50 citations)

Zooming into the lower end of the citation count range from 0 to 50, we can see there is a sharp spike at zero citations, showing that over 700,000 papers have zero citations. This sharply drops once we get to one citation and notably decreases as the citation count approaches 50. The jump at 50 is due to us capping the graph due to the large dataset, which should be noted when using “n_citations” as a measure of influence.

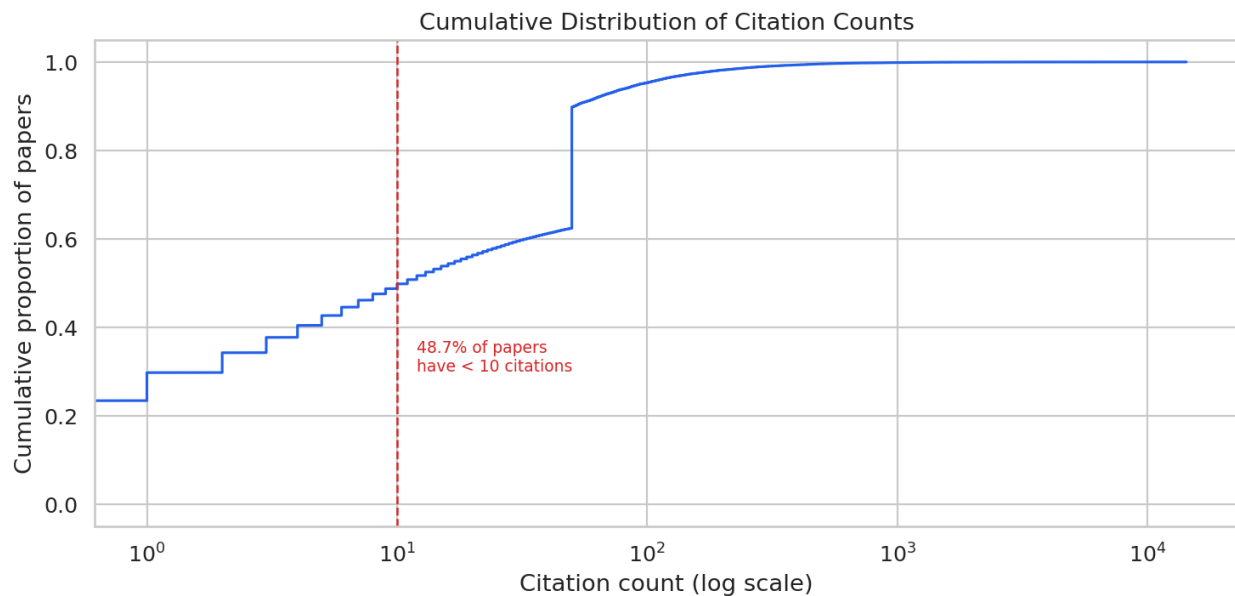


Figure 6: Cumulative Distribution of Citation Counts - 48.7% of papers have less than 10 citations

The cumulative distribution confirms the extreme concentration of citations, as about 49% of all papers have less than 10 citations. The sharp jump in the graph is around the 50 citation mark, which is consistent with the capping artifact found in figure 5. Only a small number of papers exceed 100 citations, and even a smaller portion for papers with over 1000 citations. This shows how citation count is an unreliable measure of influence.

In addition to citation count, we also analyzed how citation counts across time and publication venues.

Clustering Analysis

Since the dataset is a huge collection of research papers with no topic labels on hand for each paper, we had to figure out a different method to perform clustering. We used each paper's abstract and title, converted them into numerical features using TF-IDF, building a vocabulary of 15,000 terms fit on a 500,000-paper sample rather than the full dataset to keep memory usage manageable. We then used TruncatedSVD to compress those features down to 100 dimensions, which allowed us to avoid the large memory spikes that would have come from using standard PCA on data this large. For the actual clustering, we chose MiniBatchKMeans because it can be trained one chunk at a time, meaning we never had to load all 3 million papers into memory at once. To decide how many clusters to use, we tested several values of k and measured both inertia and silhouette score on a 100,000-paper sample before settling on a final value.

Elbow & Silhouette Inertia analysis by cluster

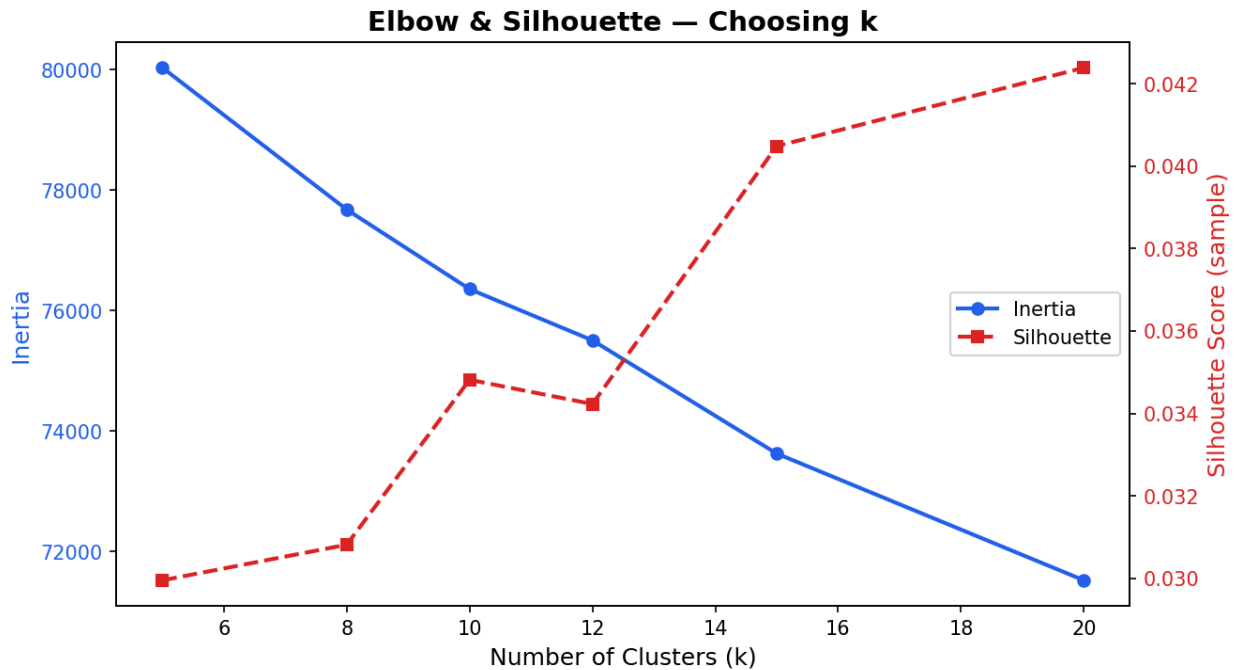


Figure 12: Elbow & Silhouette - Choosing k

The optimal number of clusters k was selected by evaluating Inertia (elbow method) and Silhouette Score across k using k -values of $\{5, 8, 10, 12, 15, 20\}$ on a 100,000-paper sample. We can see that the blue Inertia Curve decreases slowly as the number of clusters increases, while the Silhouette Score increases and decreases in sharp inclines and declines. The Inertia Curve has no sharp elbows, indicating that the data isn't forming strong, distinct clusters, and that adding more clusters isn't significantly improving the model's performance. The Silhouette Score measures how well each paper fits its assigned cluster relative to the nearest other cluster, with a value closer to 1 being more desirable. The k -value with the highest Silhouette Score was $k=20$. Since $k = 20$ gave the highest Silhouette Score among the values we tested, we chose it as a practical number of clusters, even though there isn't a clear "best" k in this case.

Median citations per cluster bar graph

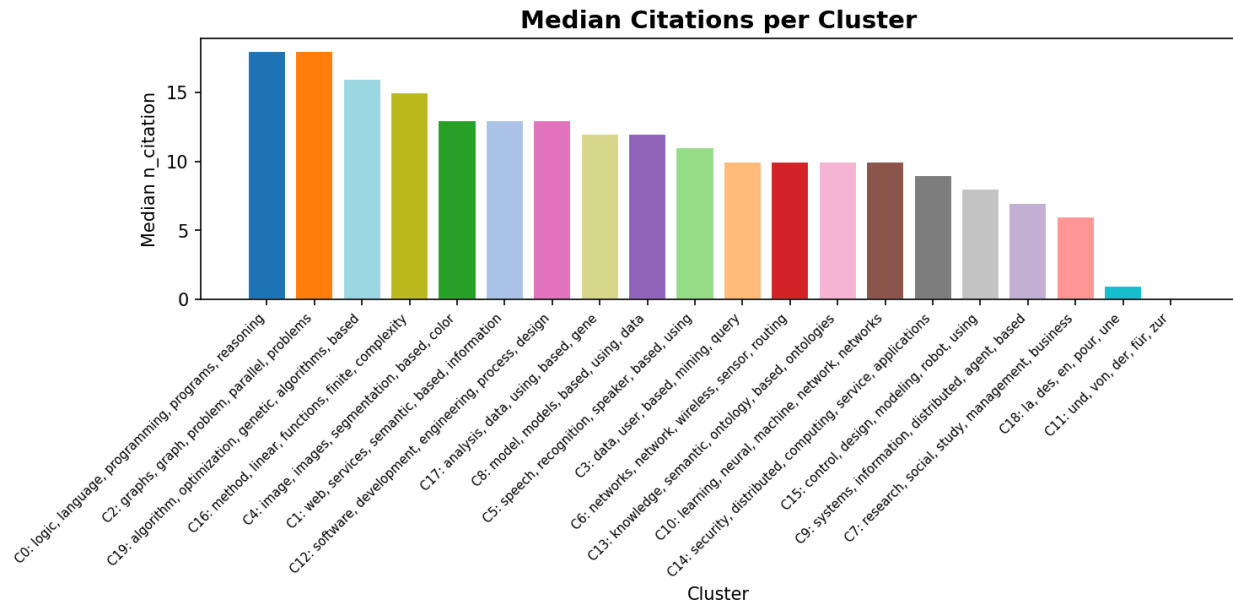


Figure 13: Median Citations per Cluster

Figure 13 shows which clusters, or research areas, are more widely cited. The least cited clusters, C18 and C11, are 2 of the only clusters using non-English words, indicating that non-English papers have limited visibility and/or are not as observed & cited by the primarily English papers in the dataset. Among the English language clusters, the bottom 3 cited are {C7: research, social, study, management, business; C9: systems, information, distributed, agent, based; C15: control, design, modeling, robot, using} and the top 3 cited are {C0: logic, language, programming, programs, reasoning; C2: graphs, graph, problem, parallel, problems; C19: algorithms, optimization, genetic, algorithms, based}. The results suggest that clusters focused on foundational or cross-functional methods such as algorithms and language tend to receive more citations, most likely because they are applicable across a wide range of research areas. In contrast, more applied or domain-specific areas, such as management, robotics, or distributed systems, get fewer citations within this dataset. This is possibly due to narrower scope or more fragmented research communities. Given the dataset's focus on computer science, broadly applicable methods are more likely to be cited across clusters, while specialized fields could remain more internally referenced.

Individual cluster growth over time

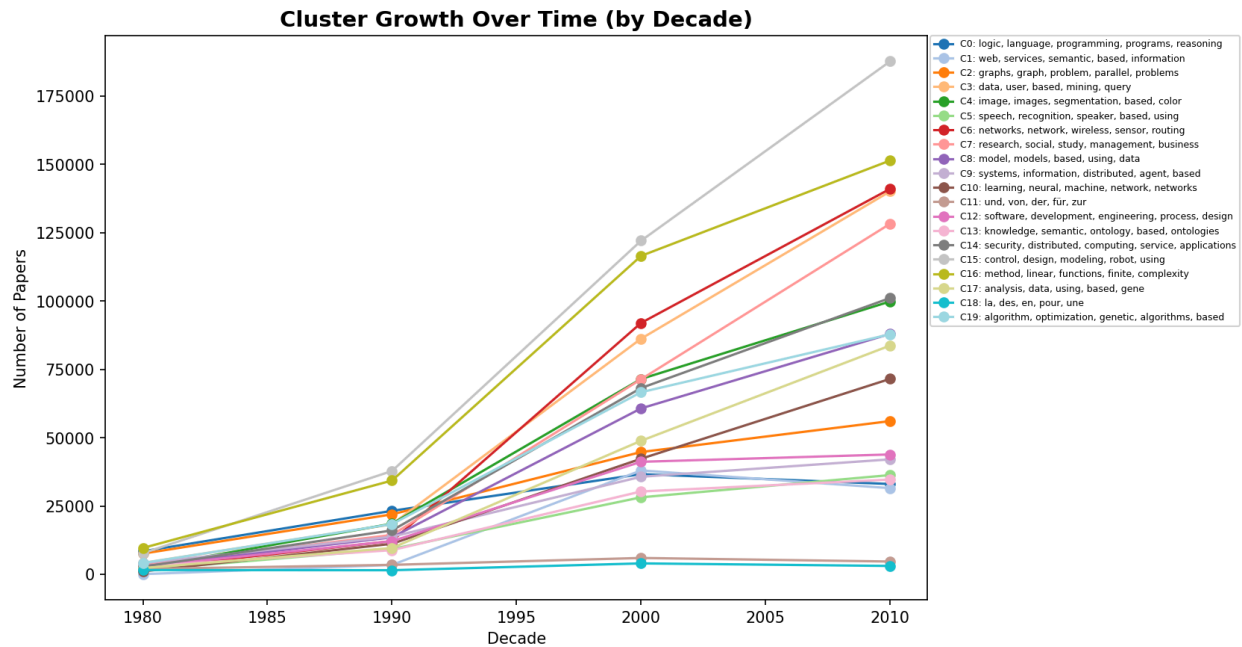


Figure 14: Cluster Growth Over Time (by Decade)

The chart tracking cluster growth in Figure 14 shows that there was a large growth in the number of papers across all clusters after 1990, and that they continued at a similar, or slightly lessened, growth rate from the years between 2000 through 2010. The top 3 clusters with the greatest number of papers by the year 2010 are {C15: control, design, modeling, robot, using; C16: method, linear, functions, finite, complexity; C6: networks, network, wireless, sensor, routing}. Something to note is that these top papers are all different from the top 3 seen previously in Figure 13 showing the most cited clusters, indicating that having more papers in a cluster might not correlate with that cluster being more widely cited. C15 has the most striking growth across the decades, showing an increasing interest in research related to modeling and robotics over time. Something interesting to note is how the number of papers in C6 has grown over time, with it being among the lowest in count around 1980, yet seeing a huge increase relative to the amount of research papers for the other clusters by the year 2000. C6's rise reflects the explosion of wireless and sensor network research in the 2000s. Overall, the chart highlights how shifting priorities in research create rapid growth in certain areas, even if those areas do not immediately appear to have higher citation numbers. This shows that research volume and citation influence capture different aspects of a field's significance, with some fast-growing domains still emerging in terms of long-term impact.

Citation Network Analysis

With the dataset cleaned and each paper assigned a cluster label, we were able to construct a directed citation graph to identify the most influential works across the entire corpus. Each paper in the dataset becomes a node in the graph, and each entry in a paper's references list becomes a directed edge pointing from the citing paper to the cited paper, so that influence flows toward frequently cited works. Since the dataset is so large, creating a graph using each paper's many citations builds something even larger, and brings us large memory challenges. To handle

this, we read the references column one chunk at a time using PyArrow's row-group API, filtered each chunk to keep only edges where both the citing and cited paper existed in our cleaned dataset, and added them to the graph incrementally rather than collecting them all at once. Once the graph was built, we computed PageRank as our primary influence metric, using a sparse matrix solver to keep memory in a manageable range. Raw in-degree, which is simply the count of how many papers cite a given work, was also computed as a secondary cross-check. Cluster labels from the clustering pipeline were joined onto each paper before any analysis began, so that every influence metric could be interpreted through the lens of the research area.

Top 20 papers by PageRank

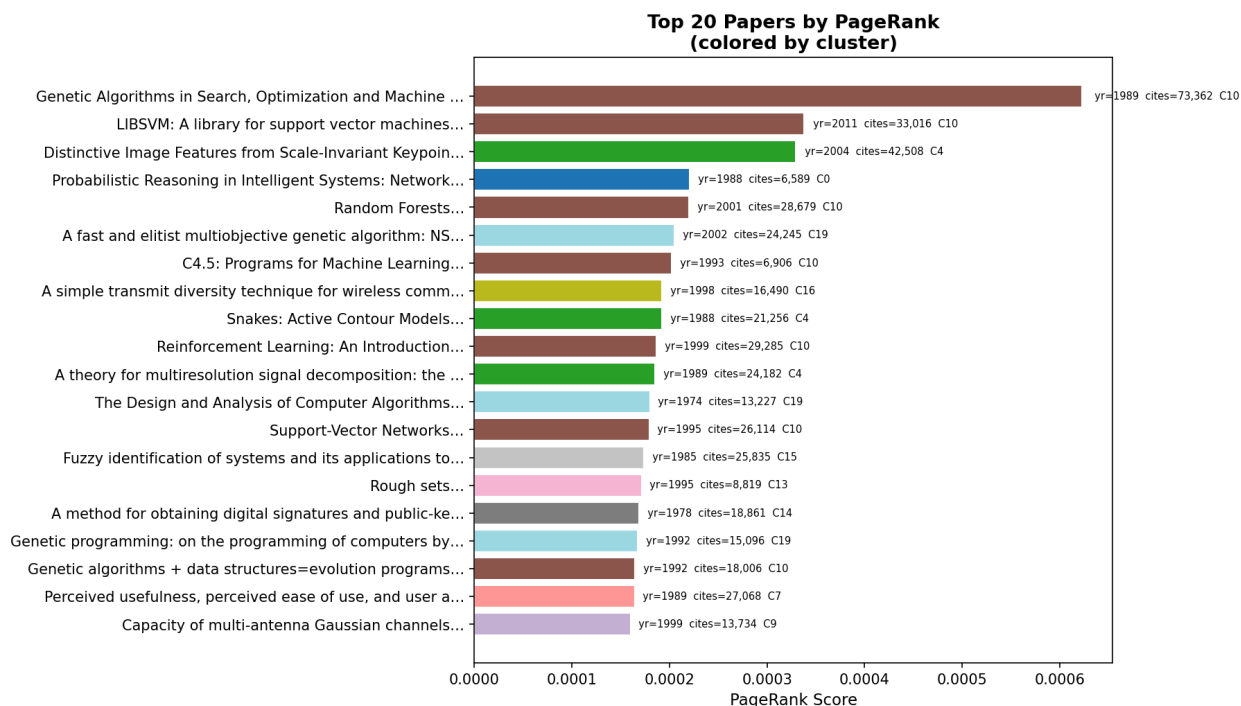


Figure 15: Top 20 Papers by PageRank

Figure 15 illustrates the most influential papers using PageRank. We are qualifying which papers are the most influential by using PageRank to not just see which paper is cited the most often, but also to see if that paper is cited by other influential works. We saw this as a more well-rounded way of measuring a paper's importance, by seeing the quality of the citations and not just the quantity. The majority of the top ranked papers come from cluster C10, with the highest ranked (by a significantly large PageRank Score compared to those below) being on Genetic Algorithms in Search, optimization and Machine Learning. This information alongside the fact that cluster C10 dominates the chart by PageRank Score shows that Machine Learning is one of the most influential research areas by PageRank.

Heatmap of citations across clusters

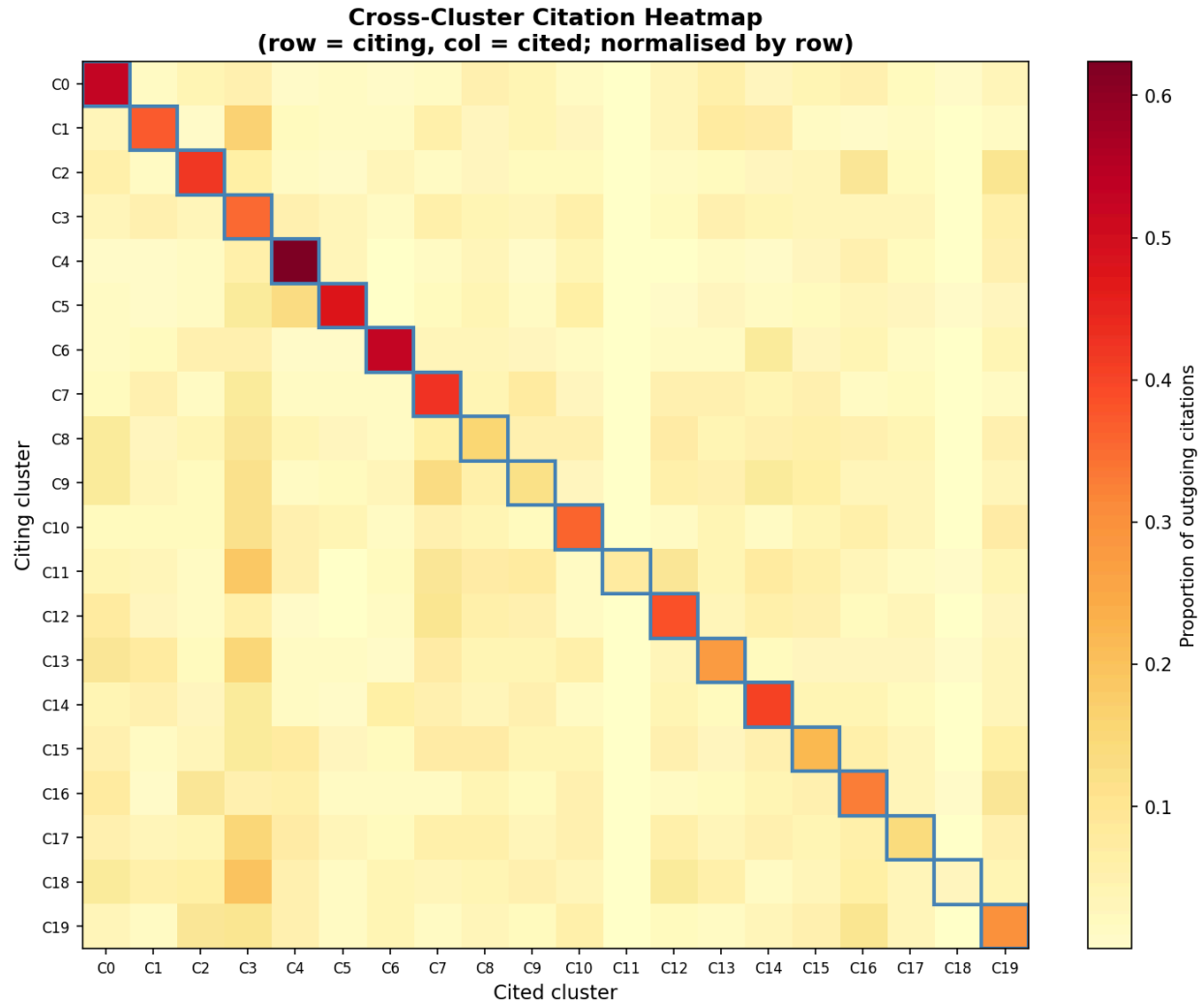


Figure 16: Cross-Cluster Citation Heatmap (row=citing, col=cited, normalized by row)

The cross-cluster citation heatmap in Figure 16 shows the proportion of each cluster's outgoing citations that go to each other cluster, with the diagonal highlighted in blue representing within-cluster self-citation rates. Overall, self-citation rates are relatively high across most clusters, meaning that papers in a given research area tend to cite other papers within that same area more than they cite outside of it. This is expected behavior for well-defined research communities that build closely on their own prior work. The most notable exceptions are C18 (la, des, en, pour, une), C11 (und, von, der, für, zur), C8 (model, models, based, using, data), and C9 (systems, information, distributed, agent, based), which all show noticeably lower diagonal values compared to the surrounding clusters. For C18 and C11, this is likely because these non-English paper clusters are too small and isolated to form strong internal citation networks of their own, so their citations scatter across other clusters rather than concentrating within. For C8 and C9, the lower self-citation rate suggests these are more generalist or interdisciplinary areas whose papers draw broadly from many different research communities rather than citing heavily within a single focused domain.

Cluster citation volume over time

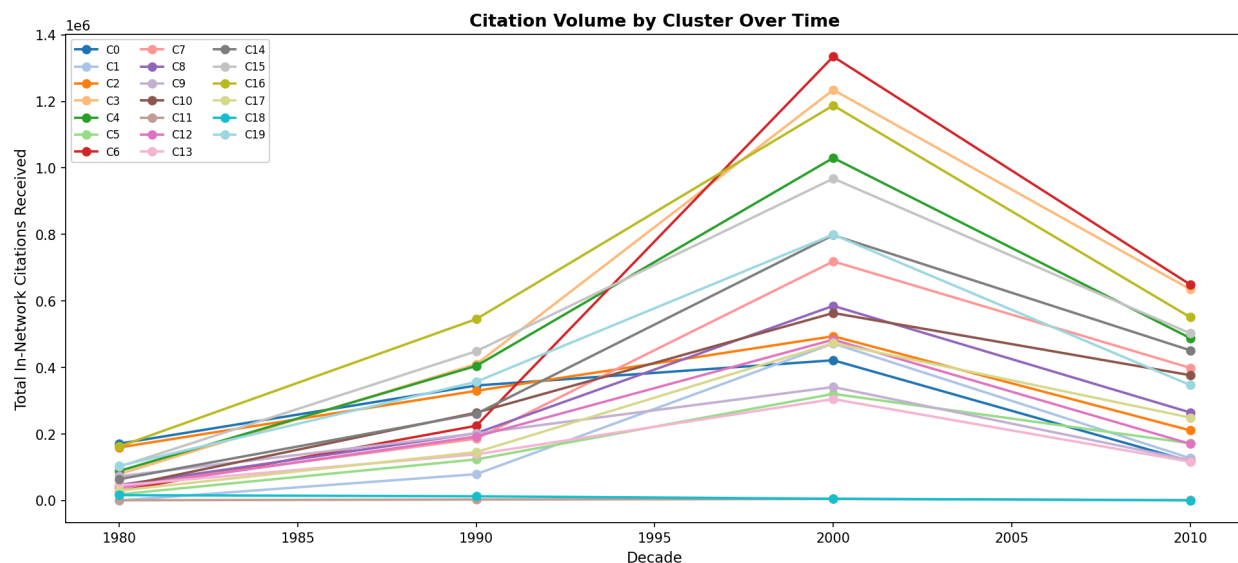


Figure 17: Citation volume by Cluster Over Time

Figure 17 shows the total in-network citations received per cluster per decade from 1980 to 2010. Citations across all clusters rise steadily through the 1990s and peak in the 2000 decade, after which they decline going into 2010. This decline is likely not a reflection of reduced research activity, but rather due to the fact that papers published in the 2010s had fewer years to accumulate citations compared to papers from the 2000s, so the 2010 data point understates what the eventual citation volume for that period would look like. The non-English clusters C18 and C11 stay flat across all decades, aligning with their isolation from the broader citation network observed in the heatmap. Among the English-language clusters, C6 (networks, network, wireless, sensor, routing) shows the most dramatic rise, peaking at approximately 1.3 million citations in the 2000 decade, reflecting the significant increase of networking and wireless research during that time. C3 (data, user, based, mining, query) and C16 (method, linear, functions, finite, complexity) also peaked high in 2000, suggesting that data-driven and mathematical methods research was generating significant citation activity during that period.

Temporal Classification Analysis

We used Temporal Classification to predict the influence that a paper published after 2009 may have using papers from before that point as training to build our prediction model. We labeled papers as influential in this model judging by the paper's citation count relative to the median for that year. If the citation count was greater than the median, it was labeled as influential, otherwise, it was not. We used this approach instead of a fixed citation threshold because it accounts for the recency bias identified in the EDA, where older papers naturally accumulate more citations over time than newer ones. To represent each paper as a feature vector, we used TF-IDF reduced to 100 SVD dimensions from the paper's title and abstract, and supplemented those text features with three structured metadata fields: publication year, reference count, and author count. TF-IDF was fitted exclusively on the training set to prevent any vocabulary from leaking into the test set. Afterwards, we compared three classifiers

(Logistic Regression, Random Forest, and LightGBM) evaluating each on accuracy, macro-F1, and training time to identify the best model for this task.

Accuracy & Macro-F1 and training time bar graphs

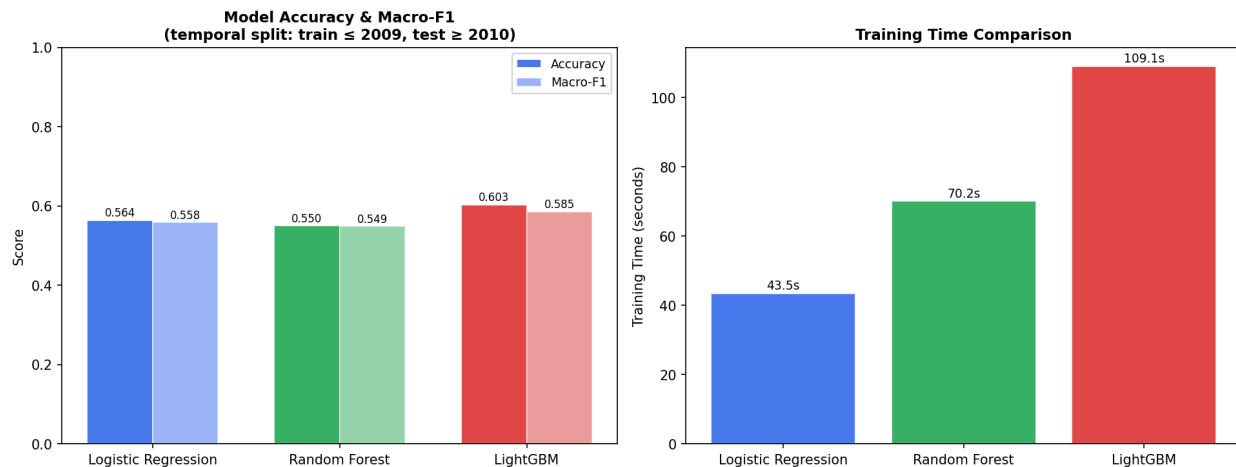


Figure 18: Model Accuracy & Macro-F1, Training Time Comparison

Figure 18 shows that the LightGBM model achieves the highest accuracy score and Macro-F1 score when compared to Logistic Regression and Random Forest. The training time for LightGBM was also significantly higher than the other models, even being over twice as long as that of Logistic Regression, which trained in 43.5 seconds compared to LightGBM's 109.1 seconds. Despite having the longest training time, we chose LightGBM as the best model given that it provided the best predictive performance with both accuracy score and Macro-F1 score. All three models scored in a similar range of 0.55 to 0.60 macro-F1, which shows the difficulty of predicting future academic influence from text and metadata alone. No single model dramatically outperforms the others, suggesting that the ceiling for this task is constrained more by the nature of the prediction problem itself than by the choice of classifier.

Feature importance graph

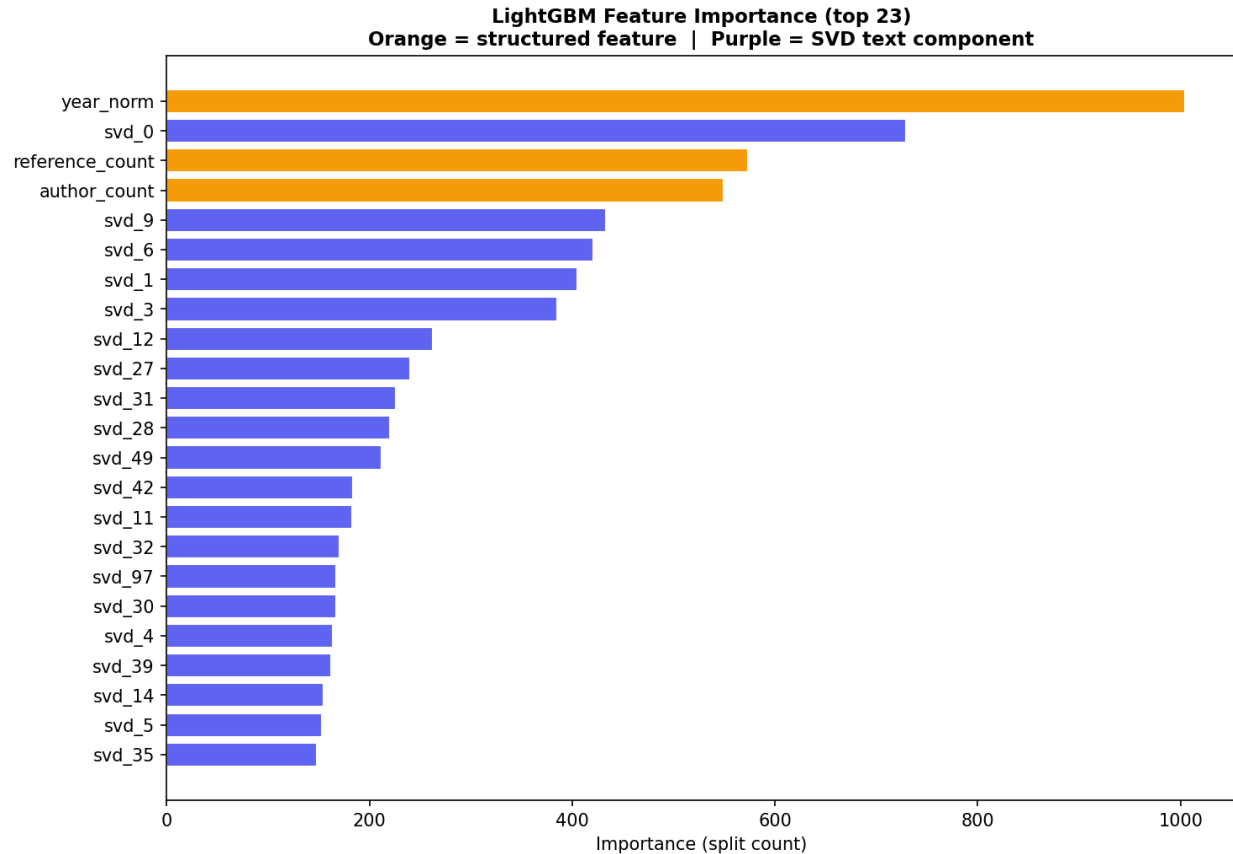


Figure 19: LightGBM Feature Importance (top 23)

Figure 19 shows which features were the most influential for the model based on the feature's split count. Year_norm, svd_0, and reference_count were the top features, with year_norm seeing a sharp increase in split count relative to the other features. This indicates that the model relies more heavily on the publication year of the article to determine its influence. Svd_0 being a highly important feature suggests that a research article covering a broad range of topics can be influential in predicting its influence, even after taking the publication year into account. reference_count and author_count also rank among the top four structured features, suggesting that papers with longer reference lists and more authors tend to be more influential. The remaining top features are all SVD text components from various dimensions, with no single dimension standing out as drastically more important than the others, indicating that influence is determined across multiple aspects of a research paper's content rather than being concentrated in one particular topic axis.

Confusion matrix for LightGBM 1565655

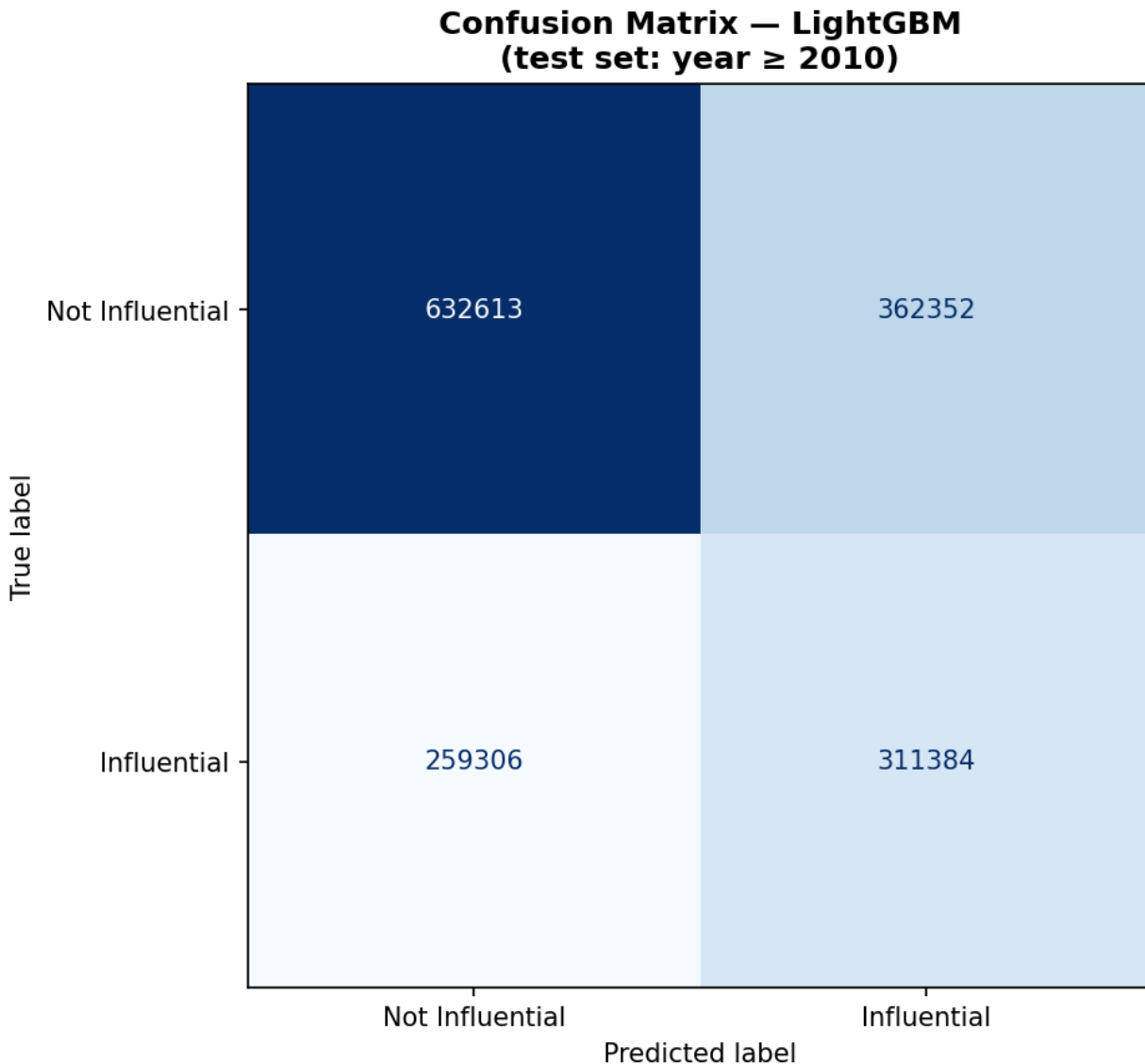


Figure 20: Confusion Matrix - LightGBM

Figure 20 shows the confusion matrix for LightGBM. The model correctly classified 632,613 non-influential papers (TN) and 311,384 influential papers (TP), while misclassifying 362,352 non-influential papers as influential (FP) and 259,306 influential papers as non-influential (FN), giving a total test set size of 1,565,655 papers. The false positive rate is $FP / (FP + TN) = 362,352 / (362,352 + 632,613) \approx 36.4\%$, and the false negative rate is $FN / (FN + TP) = 259,306 / (259,306 + 311,384) \approx 45.5\%$, meaning the model is somewhat better at identifying non-influential papers than influential ones.

Conclusion

Based on our clustering, citation network, and classification analyses, we can conclude that the DBLP dataset contains clear and meaningful structure in how research topics are

organized and how influence is distributed across computer science subjects. K-Means clustering effectively identifies primary topic boundaries and reveals that research communities predominantly cite within their own area, with foundational theoretical clusters in logic and algorithms showing the highest median influence scores. The citation network analysis confirms that structural centrality measured by PageRank provides a richer view of influence than raw citation counts alone, and that the most influential papers in the dataset are overwhelmingly foundational machine learning and algorithms papers from the 1980s through the 2000s.

Temporal classification results show that publication year and reference count are stronger predictors of influence than textual content, suggesting that a paper's eventual impact is driven more by timing and structural factors than by the specific words in its abstract. Model performance is stable across different year thresholds, indicating that the patterns learned from historical data generalize reasonably well within a decade-long window. Future work could extend this analysis with venue prestige scores as an explicit feature, author-level network centrality measures, and filtering of non-English papers prior to modeling to improve both classification performance and clustering interpretability.